DIOPHANTINE CONSISTENCY

Common remainder certificates

and explicit polynomial forms of Con(ZFC)

A quartic in 252 unrestricted integer variables,
smaller native coefficients, and improved Pell carriers

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Main result. A fixed, completely materialized polynomial

$$P_{252}(z_0, \dots, z_{251}) = \sum_{j=0}^{181} Q_j(z_0, \dots, z_{251})^2$$

has degree four and 5,340 collected monomials. Every Q_j is explicitly quadratic or affine. Composing the new arithmetic proof with the pinned direct-calculus calibration establishes

$$\mathsf{ZFC} \vdash \left[\neg\exists \mathbf{z} \in \mathbb{Z}^{252}\ P_{252}(\mathbf{z}) = 0\right] \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The new step replaces eleven separate interval certificates by one common quadratic bound on rescaled remainder coordinates. Relative to the newest matched direct parent, it removes 30 variables and ten residual equations, without increasing the degree or coefficient height.

Scope. This is an ordinary mathematical proof with exact finite algebraic verification. It does not assert a global smallest equation, a proof of consistency, or a new end-to-end proof-assistant derivation. The inherited logical and arithmetic dependencies are identified and retained in the accompanying archive.

Abstract

We construct a fixed integer quartic whose unsolvability is provably equivalent in ZFC to the syntactic consistency of ZFC. The construction uses the repository's direct numerical proof calculus, and does not encode K215 or specialize Jones's universal polynomial. Its 252 unrestricted integer variables improve the latest matched direct construction's 282. The mechanism is a general representation lemma: digit remainders with different positive divisors are rescaled to one common divisor, and their separate nonnegativity certificates are replaced by one small Euclidean norm certificate. A two-stage coefficient bound recovers the factorial digit before its factorial interpretation is available, and then recovers the ten range digits. We give constructive soundness and completeness, the full residual specification, the complete collected polynomial, and independent finite checks. Separate results complete a previously conditional radix-2^{49} native coefficient construction and lower the degrees of compact ten-positive-variable equations by reusing a smaller carrier denominator. A complemented-label encoding also permits a fully certified radix-2^{47} native tuple, with 207,214 raw coefficient bytes at the unchanged 15-variable, degree-157, 66-operation profile.

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1 What is constructed and what is being optimized

The request is for an ordinary polynomial with fixed integer coefficients, not an expression that leaves exponentiation, a proof predicate, or a consistency-dependent constant unevaluated. The principal deliverable is `direct/equation/equation.json`, whose variable order and 182 residuals define P_{252} exactly. The companion `direct/equation/expanded_polynomial.json` lists every collected coefficient. Appendix A prints the same residual specification using the fixed variables $z_0, \ldots, z_{251}$.

The relevant comparison is multidimensional. The number of variables, degree, coefficient height, expanded support, arithmetic-circuit cost, and length of the coefficient specification are different measures. Reducing one can increase another. The repository already contains equations with fewer variables than 252 on other routes and at other degrees; this report does not erase those alternatives. Its principal improvement is within the direct proof-calculus quartic route, which has small literal coefficients and needs neither a native machine tuple nor a universal-polynomial specialization.

All comparisons use the fixed repository commit `55d1f72aedebf892e446c90d35fa1ae37689b0f2`. The collection index, active constructions, and newest matched incoming parents were inspected. In particular, the direct matched parent is the `strict_dimension_centered` construction in the incoming *Cofinal Anchors and Height One, Round 17* archive [2]. Comparing only against the older collection index would miss this parent.

Table 1: Matched direct quartics over unrestricted integers. Coefficient height is the maximum absolute collected coefficient.

Measure	Pinned parent	New P_{252}
Variables	282	**252**
Quadratic or affine residuals	192	**182**
Total degree	4	4
Nonzero collected monomials	5,491	**5,340**
Coefficient height	128,446	128,446

Thus the new construction saves $30/282 \approx 10.64\%$ of the variables, $10/192 \approx 5.21\%$ of the residuals, and $151/5491 \approx 2.75\%$ of the expanded monomials. These are finite counts of the supplied objects, not estimates from a symbolic template.

The general DPRM theorem explains why a Diophantine representation of a recursively enumerable refutation predicate exists [6]. It does not supply these particular small explicit coefficients. Likewise, Jones's variable reduction and later universal-pair work provide important comparison frameworks [7, 8]; they are not substituted for the direct construction or its calibration here.

2 A common certificate for many remainders

For integers a, b, c, define

$$G(a, b, c) = a^2 + b^2 + c(c + 1).$$

This expression is always nonnegative. Conversely, every $n \in \mathbb{N}$ has such a representation. Indeed, $4n + 1$ is not of the form $4^u(8v + 7)$, so the three-square theorem represents it by three squares. Reduction modulo four shows that exactly one of those squares is odd. Writing the odd root as $2c + 1$ and the other roots as $2a, 2b$ gives $n = G(a, b, c)$. This is the same elementary integer-sign representation used by the parent.

Lemma 2.1 (Integral rescaling of a remainder)**.** *Let L, M be positive integers with $L \mid M$, and put $v = M/L$. For integer U, Q, R, the equation*

$$vU = MQ + R$$

implies $v \mid R$. The integer $r = R/v$ then satisfies $U = LQ + r$. Conversely every integer solution of the latter equation gives a solution of the former by $R = vr$.

Proof. Both vU and $MQ = LvQ$ are divisible by v, so their difference R is divisible by v. Division by the nonzero integer v gives the assertion. The converse is direct multiplication. In the polynomial system v is an integer coordinate constrained by $Lv = M$; no rational coordinate is introduced. □

Theorem 2.2 (One norm certificate for a family of small remainders)**.** *Let $M, L_1, \ldots, L_s$ be positive integers with $L_i \mid M$, and $v_i = M/L_i$. Suppose the scaled digit equations*

$$v_i U_i = MQ_i + R_i \qquad (1 \le i \le s)$$

hold, together with

$$M^2 = 1 + 4\sum_{i=1}^{s} R_i^2 + G(a, b, c). \tag{1}$$

Then $r_i = R_i/v_i$ is an integer with $|r_i| < L_i/2$. If the ordinary least nonnegative remainder r_i^0 of U_i modulo L_i satisfies $r_i^0 < L_i/2$, then $r_i = r_i^0$.

Conversely, if $s \le 11$ and an ordinary digit family has $0 \le r_i^0 < L_i/32$, there exist $a, b, c \in \mathbb{Z}$ for which its scaled remainders $R_i = v_i r_i^0$ satisfy (1).

Proof. Lemma 2.1 gives integrality. Nonnegativity of G implies $4R_i^2 < M^2$, hence $|r_i| < L_i/2$. The difference $r_i - r_i^0$ is a multiple of L_i and has absolute value strictly less than L_i; it is therefore zero.

For the converse, $0 \le R_i < M/32$, and so

$$4\sum_i R_i^2 < \frac{44}{1024} M^2 = \frac{11}{256} M^2 < M^2.$$

The difference $M^2 - 4\sum_i R_i^2$ is a positive integer, so subtracting one leaves a nonnegative integer. Represent it by G. □

The completeness hypothesis is essential: a common norm need not accept every arbitrary tuple of remainders in their individual intervals. Here it accepts the much smaller canonical remainders produced by the binomial constructions. The next two sections prove that margin without assuming the conclusion of the factorial step.

Remark 2.3 (A valid smaller coefficient). The coefficient four in (1) can be replaced by two when the $1/32$ canonical bound is available in both directions. The new bound is $|r_i| < L_i/\sqrt{2} < 3L_i/4$. Thus $|r_i - r_i^0| < 25L_i/32 < L_i$, which still forces equality. Completeness uses $22M^2/1024 < M^2$. The released primary polynomial uses four for the particularly transparent centered interval. No second numerical endpoint is claimed from this observation.

3 Applying the lemma to the direct arithmetic source

The inherited source uses eighteen certified power requests. These are implemented by ordinary polynomial Pell relations. A symbol such as q^t below denotes the integer value of an existing output coordinate after its meaning has been proved; it is not an operation left inside the final polynomial.

The eleven replaced remainder intervals have only four different denominators:

Denominator	Number	Purpose
q^t	2	Factorial binomial; encoded index
q^{t+1}	5	Four syntax representatives; derived label
q^{y+1}	2	Two pending-multiset representatives
q^x	2	Two beta quotients

The parent proves

$$t \geq 16, \quad y \geq 1, \quad x \geq 2 + (t+1)y$$

before any of these eleven interval guards is used. Set $M = q^x$ and introduce three ratio coordinates by

$$q^t v_t = M, \qquad q^{t+1} v_{t+1} = M, \qquad q^{y+1} v_y = M. \tag{2}$$

All the divisors in these equations are positive powers of the same positive base, with exponents at most x, so the ratios are positive integers. For the fourth denominator the ratio is simply one.

The factorial digit formerly had the form

$$U = q^t(B + q\omega) + r.$$

It becomes

$$v_t U = M(B + q\omega) + R.$$

The quantity $q\omega$ already has a circuit coordinate, so this is a quadratic residual in the actual coordinates. Each range digit formerly had the form

$$U = Nq^k\omega + r$$

and becomes $v_k U = NM\omega + R$. The four separate product coordinates Nq^k are consequently replaced by one shared coordinate NM. The new remainder coordinates occur only in their scaled digit equations and the one norm equation; they do not change a private Pell frame atom or a requested power.

3.1 The order in which signs and power meanings are recovered

The new proof retains the parent's signed-frame and semantic-anchor arguments, followed by its coordinate-orientation and strict-budget arguments. Their conclusions are used in the following order:

1. The three retained initial sign predicates imply $B \geq 1$ and $q \geq B + 1 \geq 2$. The cofinal horizon has $t \geq 16$.
2. Signed Pell orientation and the request at exponent t identify $V_0 = (F^2 + 1)^t$. The raw ceiling $FB = V_0 + s$ with $s \geq 0$ then gives $F \geq 1$ and $B > F$, hence $B \geq 2$.
3. The beta value $y = V_0(V_0 - 1)$ and the two endpoint equations give $x \geq 2 + (t+1)y \geq 19$.

4. The requests at base $q+1$ exclude negative actual representative indices. Thus the nine actual CRT representatives are nonnegative. Writing $H = 1 + \sum a_i$ gives $H \geq 1$.
5. The remaining requests identify $N = B^x$ and $q = (BH)^x = B^x H^x$. Every denominator and numerator power is now interpreted.

Only after these steps do we interpret (2) and the common norm. In particular, no deleted remainder sign is needed to establish the positive divisor or power used to recover that sign. The inherited proofs are included under `direct/upstream/round17/`; their boundary is discussed in Section 6.

4 The factorial digit is recovered before assuming $F = t!$

Lemma 4.1 (An elementary coefficient bound)**.** *For integers $m, t \geq 1$ and $0 \leq j \leq t$,*

$$\binom{m+t}{j} \leq (m+1)^t.$$

Proof. For $t = 1$ the assertion is immediate. In the next Pascal row, the coefficients with $1 \leq j \leq t$ are bounded by $2(m+1)^t \leq (m+1)^{t+1}$; the constant edge is one. The newly included last coefficient satisfies

$$\binom{m+t+1}{t+1} = \prod_{i=1}^{t+1} \left(1 + \frac{m}{i}\right) \leq (m+1)^{t+1}.$$

This completes the induction. □

Apply the lemma with $m = F^2$. The already interpreted factorial numerator is $U = (q+1)^{F^2+t}$, and every binomial coefficient through position t is at most $V_0 = (F^2+1)^t$. The raw ceiling, before factorial semantics, gives

$$V_0 \leq BF < B^2.$$

Because $q = B^x H^x$ and $x \geq 8$,

$$\frac{V_0}{q-1} < \frac{B^2}{q-1} < \frac{2}{B^{x-2}} \leq \frac{1}{32}. \tag{3}$$

The true low part in the binomial expansion is

$$r^0 = \sum_{j=0}^{t-1} \binom{F^2+t}{j} q^j.$$

The geometric-series bound and (3) give

$$0 \leq r^0 < \frac{V_0 q^t}{q-1} < \frac{q^t}{32}.$$

In particular this is indeed the ordinary remainder modulo q^t. Theorem 2.2 reconstructs it exactly from the new signed remainder. Dividing the restored digit equation by q^t and reducing modulo q gives

$$B \equiv \binom{F^2+t}{t} \pmod{q}.$$

Both quantities lie in $[0, q)$: $B < q$ follows from the retained initial output slack, and the binomial coefficient is at most $V_0 < q$. Consequently

$$B = \binom{F^2 + t}{t}.$$

The unchanged factorial fixed-point theorem of the direct source now gives $F = t!$. The proof above has not used this identity early.

5 The remaining range digits and constructive completeness

After the factorial interpretation, the inherited arithmetic congruence $B \mid C_0 + F^2 + 1$ excludes a zero encoded-index representative: the factorial formula gives $B > F^2 + 1$. Thus $C_0 \geq 1$ and $H \geq 2$.

For an ordinary nonnegative upper representative a and a relevant position $0 \leq j \leq k \leq x$,

$$\binom{a}{j} \leq \max(1, a)^k \leq H^x.$$

This includes $a = j = 0$; using a^k without the maximum would give a false bound at that edge. For the derived label h, the inherited literal degree bound is $h \leq H^\ell$ with $\ell = 4,194,304$, and the factorial/endpoint estimate gives $\ell(t + 1) < x$. Its relevant coefficients therefore have the same H^x upper bound.

Since $q = B^x H^x$,

$$\frac{H^x}{q - 1} < \frac{2}{B^x} \leq \frac{1}{128} < \frac{1}{32}.$$

The same geometric-series argument now proves $0 \leq r^0 < q^k/32$ for each of the ten range digits. The common norm therefore recovers every ordinary remainder and its intended quotient. Restoring each deleted interval slack by $s_i = q^k - r_i^0 - 1$ gives $s_i \geq 0$; a fresh G-representation supplies the former product sign certificate if it is wanted. The unchanged divisibility and prime-power range-decoding proof applies.

For completeness, start with a finite refutation and use the parent's cofinal horizon, beta-CRT construction, amplified CRT representatives, even dominant semantic anchor, and private Pell witnesses. These genuine witnesses satisfy both coefficient estimates just proved. Set $v_i = M/L_i$ and $R_i = v_i r_i^0$. The three ratio equations and eleven scaled digits hold. Every R_i is below $M/32$, so Theorem 2.2 supplies the three norm witnesses. Replace the four old denominator products by NM, retain all other semantic data, and apply the supplied integral elimination and translation maps. This produces a zero of the final polynomial from any finite refutation.

6 The explicit quartic and its logical calibration

Theorem 6.1 (The released direct equation)**.** *Let $Q_0, \ldots, Q_{181} \in \mathbb{Z}[z_0, \ldots, z_{251}]$ be the fixed residuals printed in Appendix A and serialized in* `direct/equation/equation.json`*. Define*

$$P_{252} = \sum_{j=0}^{181} Q_j^2.$$

Then P_{252} has degree four, 5,340 collected monomials, and coefficient height 128,446. Using the pinned direct-calculus calibration,

$$\mathsf{ZFC} \vdash (\exists \mathbf{z} \in \mathbb{Z}^{252}\ P_{252}(\mathbf{z}) = 0) \longleftrightarrow (\exists p\ \mathrm{Refutation}_{\mathsf{ZFC}}(p)).$$

Therefore ZFC proves unsolvability of this fixed equation equivalent to $\mathrm{Con}(\mathsf{ZFC})$.

Proof. An integer sum of squares is zero exactly when every summand is zero. The literal Horn factoring, integral monic projections, and integer translations preserve existential solvability, with explicit inverse maps. The preceding sections prove the two directions between the new arithmetic source and the parent's interpreted direct arithmetic source. The latter decodes a finite refutation in the numerical calculus C222. The retained endpoint interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their composition gives the displayed refutation equivalence. Negating it gives the consistency formulation. The degree and coefficient counts follow by exact collection of the supplied residual squares. □

The theorem uses the repository's existing ordinary arithmetic and logical calibration. The new contribution is the remainder representation and its two-stage recovery proof, together with the exact polynomial transformation. The archive preserves the relevant inherited proof texts, source hashes, and calibration names. It does not claim that a Python coefficient audit proves an unbounded arithmetic theorem, nor that the whole transitive Lean/Rocq development was rebuilt in this iteration.

No consistency assumption is used to prove the equivalence. Under consistency, the equation has no integer zero; if a syntactic contradiction exists, the constructive direction produces a zero. Neither alternative is decided by this report.

6.1 Why precisely thirty variables disappear

Eleven separate interval predicates previously needed three ternary sign witnesses each, for 33 variables. The new construction adds three common divisor ratios and three witnesses for the one norm predicate. This saves 27. Replacing the four products Nq^k by NM saves three more. The old interval slacks were already removable by affine elimination; they are not counted again. The final literal count is consequently $282 - 27 - 3 = 252$.

7 A smaller denominator in the compact Pell carrier

This secondary result improves the ten-positive-variable equations in the incoming *Certificate Reductions* package [3]. It is a separate K215/native route. Its constants are the complete fixed indexed tuple of that package, not the direct quartic's small coefficients.

The predecessor's first Pell base is $P_0 = 2XY^2 + 1$, with $X = wq$ and $Y = q^3 s$. It uses a carrier denominator $D = Y^2$. The following proposition identifies the actual denominator requirements.

Proposition 7.1 (Carrier substitution)**.** *Keep the predecessor's first Pell base and all its definitions other than the carrier denominator. Any positive polynomial value D satisfying*

$$D > 2, \qquad q^4 \mid D, \qquad D \mid 4XY^2$$

on every positive input can replace Y^2 in the carrier without changing its equivalence with the native refutation condition.

Proof. The unchanged product-square extraction makes the rational carrier an integer and yields the same main and first Pell representations. Its sign transport uses only $D > 2$. It therefore gives the same positive outer data and $R < q^4$. Clearing the integral carrier gives $D \mid hHe$, with $h = q - 1$ and $H \equiv -1 \pmod{q}$. Since $q^4 \mid D$ and $\gcd(q, hH) = 1$, we obtain $q^4 \mid e$.

The first Pell congruence is $k \equiv 2n_0 \pmod{4XY^2}$, and the unchanged growth argument gives $1 \le n_0 < R$. Thus

$$q^4 \mid 2n_0 - R - 1, \qquad |2n_0 - R - 1| < R < q^4,$$

so $2n_0 = R + 1$. Returning to the full congruence gives $4XY^2 \mid e$, hence $D \mid e$. The rest of the projection and native decoding is unchanged.

Conversely, canonical first Pell witnesses give e a positive multiple of $4XY^2$, so e/D is a positive integer. The predecessor's fresh strong and auxiliary witnesses make the padding coordinate $n = gy + Wt + e/D - \kappa$ positive, because $g > \kappa$ and the other summands have the stated nonnegative signs. The carrier equation holds. □

Set $D = XY$. Both divisibilities are immediate, and the evaluation can compute $D = XY$, then $V_0 = DY$, in place of Y^2, then $V_0 = XY^2$. The number of multiplication gates is unchanged. The denominator degree falls from eight to six, lowering the final degree by four.

Carrier	Positive variables	Binary operations	Degree
Parent $D = Y^2$, weight V^2	10	78	630
New $D = XY$, weight V^2	10	78	626
Parent $D = Y^2$, weight bV	10	79	468
New $D = XY$, weight bV	10	79	464
New $D = q^4$, weight bV	10	80	460

The full formulas, literal division-free DAGs, and the nonzero leading homogeneous forms are in `compact/`. The complete predecessor proof is retained. These are modest degree improvements, not a claim to a new universal variable bound. The 79/622 variant from $D = q^4$ and weight V^2 is dominated by 79/464 and is not promoted.

8 Native coefficients: a completed controller and a smaller radix

The direct quartic is the principal result of this report. A separate native route keeps fifteen positive integer variables and a 66-operation polynomial of degree 157, while reducing the size of its four fixed integer coefficients. This section separates a completed prior proposal from a new change to the encoding.

8.1 The polynomial is fixed before the controller is chosen

The fifteen independent positive variables are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma.$$

For a specified whole coefficient tuple (β, K_0, M_C, M_F), define the following polynomial abbreviations:

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, \\ X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, \\ c &= kY + \eta, & \ell &= EkY, \\ a &= E + Y, & H &= 4a + 3, \\ \Delta &= a^2 + H, & D &= X + ac + \gamma H, \\ V &= f^2(cT - R) - c, & S &= i\Delta c^2. \end{aligned}$$

$$R = [(q-1)(q-F) + \alpha](q^2 - 1) + (M_C + qM_F)J.$$

Put

$$\begin{aligned} N_0 &= \tau^2 - \ell(\ell + k), & N_1 &= D^2 - \Delta c^2, \\ N_a &= S^2(V^2 - y^2) + y^2, & N_k &= k - hE - R, \\ N_t &= (K_0 + w)C + (q - F) - z(q-1), & N_s &= \Delta f^2 - S^2. \end{aligned}$$

The single equation is

$$\boxed{N_0 N_1 N_a N_k N_t N_s - \Delta = 0.} \tag{4}$$

Its exact straight-line circuit has 38 multiplications, 15 additions, and 13 subtractions. The apparent top terms of N_1 cancel: with $Z = X + \gamma H$, $N_1 = 2acZ + Z^2 - Hc^2$. The six factor degrees are therefore $22, 18, 62, 7, 2, 46$. Setting all inputs equal to an indeterminate t gives leading term $2^{17}\beta^{91}t^{157}$, so the degree is exactly 157. The four named constants are not quantified unknowns. The controller prime below is not part of the coefficient tuple; it is an explicit witness used in the completeness proof.

8.2 Closing the conditional radix-2^{49} construction

The repository's shared-marker radix49 proposal [4] already contains the encoding and its ordinary native/Pell proof. Its remaining conditions were finite controller conditions. We complete precisely those conditions and materialize the entire coherent tuple.

Let $\rho = 2^{49}$, $L = 8807$, $B = \rho^L$, and $d = 49L = 431543$. Choose the fixed prime

$$\lambda = 434011 > d, \qquad \lambda - 1 = 1887 \cdot 230.$$

For $H_{\mathrm{ctrl}} = 1 + BK_0 + B^2$, exact computation gives:

Modulus	B	K_0	H_{ctrl}
7	4	5	2
49	18	47	44
8807	1901	973	3135
434011	398487	335571	6522

The exact orders are

$$\mathrm{ord}_{49}(B) = 3, \quad \mathrm{ord}_{8807}(B) = 629, \quad \mathrm{ord}_{434011}(B) = 18870.$$

The required valuation-one lift is

$$B^{\lambda-1} \equiv 96356084144 = 1 + 222013\lambda \pmod{\lambda^2}.$$

Since $0 < 222013 < \lambda$, the lift has valuation exactly one. The full modulus 49 is checked explicitly, rather than only its prime divisor seven. Deterministic trial division proves the asserted primes; annihilating powers and every maximal proper-power test prove the orders.

These facts discharge all the finite premises of the inherited conditional theorem. Its carry, phase-alignment, safe-dummy, rank-tuning, Pell, and refutation arguments then give the ZFC-provable equivalence between positive unsolvability of (4) at this tuple and Con(ZFC). The new work here is the finite completion and complete coefficient specialization; the shared-marker architecture is prior work.

The materialized radix49 tuple has bit lengths $(431543, 433602, 431543, 431544)$ in the port order (β, K_0, M_C, M_F), totaling 216,030 raw unsigned bytes. Its four deterministic gzip files total 8,774 bytes. The corresponding matched radix53 and radix55 tuples occupy 233,666 and 242,484 raw bytes. Raw coefficient storage, compressed storage, and printed-page length are different metrics. The complete coefficient recipe, binary files, standalone verifier, and inherited proof texts are supplied under `radix49/`.

8.3 Complemented labels permit a radix-2^{47} encoding

The radix49 completion suggests a further change to the coefficient construction. The previous radix47 local obstruction [5] concerned unchanged selector coefficients: its predecoded carry allowance was too large to infer that a cell selected at most one tile. Instead of discarding that obligation, change the numerical labels so the full coefficient sum becomes smaller.

For each original tile (w_i, n_i, p_i, s_i), apply the bijections

$$w_i' = 255 - w_i, \quad p_i' = 255 - p_i, \quad n_i' = 511 - n_i, \quad s_i' = 511 - s_i.$$

The west/east labels use the same eight-bit bijection and the north/south labels the same nine-bit bijection. Thus edge equality and inequality, including the typed Start and forbidden-color conditions, are preserved. The seven physical marker bits and the occupancy bits are kept. An empty cell still has zero physical label bits; complementation is the relabeling of genuine tiles, not an unconditional complement of every empty bit.

Put

$$u_i = w_i + 2^8 n_i + 2^{17} p_i + 2^{25} s_i.$$

The complemented packed label is $2^{34} - 1 - u_i$, so the new selector coefficient is

$$c_i = 1 + 2^{10}(u_i + 1) \qquad (0 \le i \le 942). \tag{5}$$

Every nonselector coefficient and physical support position is as in the shared-marker construction. Exact reconstruction of all 943 tiles gives

$$\sum_e c_e = 3,953,059,820,717,999, \qquad \max_i c_i = 15,049,678,914,561 < 2^{44}.$$

The old complete coefficient sum was 12,917,846,596,842,415. Both sums include the nonselector controls; the reduction is not obtained by omitting any coefficient from the proof.

The full new coefficient tuple

Now set $\rho = 2^{47}$, $L = 8807$, $B = \rho^L = 2^{413929}$, $\beta = B - 1$, and $\Lambda = \rho^{4393} + \rho^{5495}$. Besides (5), the complete nonzero controls are

$$\begin{aligned} c_{1023+j} &= 2^{10+j}, & c_{1003+j} &= 2^{27+j} & (0 \le j \le 7),\\ c_{1013+j} &= 2^{18+j}, & c_{983+j} &= 2^{35+j} & (0 \le j \le 8),\\ c_e &= 2^{44} & & & e \in \{992, 1011, 1022, 1031, 1095, 2195, 3296\}. \end{aligned}$$

All other c_e vanish. Let $\mathcal{E}$ be their support together with the two coefficient-zero dummy positions 1063 and 1064. It has 986 positions. Define

$$\begin{aligned} K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} + \sum_{e=0}^{3296} c_e(\rho^{4393-e} + \rho^{5495-e}),\\ M_C &= \beta - \sum_{e \in \mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063},\\ C_{\text{core}} &= \sum_{e=1013}^{1022} \rho^e + \sum_{e=1065}^{1073} \rho^e + \rho^{2195} + \rho^{3296},\\ M_F &= 2B + 3 + (2^{47} - 2)\Lambda + C_{\text{core}} - \rho^{8792} + (2^{28} - 3)\rho^{7694} - \rho^{7691}. \end{aligned}$$

These are fixed integers. Every exponent and finite sum limit is fixed. The exact 943-tile data and the complete 3297-slot coefficient array are supplied; all four binary integers are also materialized. In particular, no unspecified tile enumeration or proof search is needed to define them.

The positive native high-mask padding behind the compressed M_F formula is fifteen full fields at 8792–8806, three full fields at 7691–7693, and bits 2–27 at field 7694. Its population is $15{\cdot}47{+}3{\cdot}47{+}26 = 872$. Together with the unchanged 114 core tests, this gives 986 ones, exactly matching the physical low support. The source shift by β is applied once. All four ports are changed coherently; none is imported numerically from the radix49 tuple.

Why the old carry obstruction no longer applies

The complete predecoded convolution bound, including the exceptional dummy and every temporal phase, maps the carry interval $0, \dots, 29$ into itself. Thus

$$\#\{\text{selected tiles}\} + k \le 943 + 29 = 972 < 1024.$$

The ten low tested bits therefore recover zero or one selected tile. The remaining payload bits recover the complemented label and, by the same shared-marker counting argument, all seven marker bits individually.

After this decoding, the selector maximum is below $\rho/8$ and the entire nonselector control sum is $\rho - 1024$. The complete decoded convolution and temporal estimate, even with the earlier incoming carry, is strictly below 13ρ. Hence the actual outgoing carries are at most 12. This bound protects the anchor predecessors and all subsequent local resets.

At radix47 an occupied clause has raw value $\rho + 1$. Its canonical digit is one and it emits carry one. This carry must not be silently dropped. The proof explicitly resets it before every later tested copy, anchor, padding, or origin field. For example, the larger clause predecessor has raw bound at most $1 + 7 \cdot 2^{44} - 2^{10}$, and adding 12 still leaves it below ρ. The two phase-anchor predecessor identities

also leave room for every temporal phase plus carry 12. They recover integral temporal phase and whole-cell alignment as before. The raw padding predecessors at 7690 and 8791 are bounded by a single selector coefficient plus the incoming carry, so both reset. The complete zero fields 7691–7695 and 8792–8806 then validate the new padding.

These arguments preserve the positive-source slack, all simultaneous safe-dummy changes, and the actual rank identity. They supply the same canonical $R \equiv 3 \pmod 4$ and $\mathrm{pc}(R) = 3dN + 2$ interface required by the P66 converse. The full ordinary proof and the independent review are included in `radix47/`.

The new finite controller

Here $d = 47 \cdot 8807 = 413929$. Choose

$$\lambda = 1012691 > d, \qquad \lambda - 1 = 10 \cdot 101269, \qquad 101269 = \mathrm{lcm}(23, 4403).$$

The exact residues are:

Modulus	B	K_0	$1 + BK_0 + B^2$
47	12	5	17
8807	2677	4079	5042
1012691	959313	1005855	834350

The exact orders at these three moduli are respectively 23, 4403, and 1012690. Finally,

$$B^{\lambda-1} \equiv 1 + 282338\lambda \pmod{\lambda^2}.$$

All controller units are nonzero and the lift has valuation one. The same cofinal CRT and Boolean-subset rank-tuning proof now applies with the new d, prime, spacing, and tuple. Together with the complete carry/alignment proof, it establishes the ZFC-provable equivalence between positive unsolvability of (4) at this tuple and Con(ZFC).

Tuple	β bits	K_0 bits	M_C bits	M_F bits	Raw bytes
Radix55	484385	486696	484385	484386	242484
Radix53	466771	468998	466771	466772	233666
Completed radix49	431543	433602	431543	431544	216030
New radix47	413929	415904	413929	413930	**207214**

The new tuple saves 26,452 raw bytes, about 11.32%, against the verified radix53 tuple. Its compressed four-coefficient payload is 8,356 bytes. The scalar $B = 2^{413929}$ is also smaller than the collection's previously stated 2^{429959} native-base endpoint; this statement concerns that scalar, not a comparison of every metric of the two different encodings.

Label complementation itself has predecessors in the repository. The new result is this complete indexed943/shared-marker/radix47 composition, the corrected carry bounds and resets, its entire coherent tuple, and the completed controller. It is not a claim to have invented complementation or to have proved a globally smallest coefficient tuple.

9 Verification, reproducibility, and limits

The final archive includes complete defining objects, not just programs that might eventually find their coefficients. The direct equation can be read directly from the residual JSON or the printed appendix. Its expanded JSON contains the whole collected quartic. No search for a solution or consistency-dependent computation is part of coefficient generation.

The direct finite audit checks the exact deletion and replacement of source rows, all remaining guards, every ratio equation, every scaled digit, the shared-product aliases, all inherited Horn factoring edits, 205 monic eliminations with restoration polynomials, and the final integer translation. It independently recollects every coefficient of the 5,340-term polynomial. Deliberately altered ratio, digit, norm, restoration, translation, and expanded-coefficient data are rejected. Normal Python and optimized Python both retain the checks.

The finite native audits independently reconstruct complete coefficient tuples, check prime/order/lift certificates and controller units, and verify the literal P66 circuit and its degree. Exact integer arithmetic is used throughout. These checks supplement the ordinary carry, Pell, CRT, and refutation proofs; they do not replace them.

For the direct component, run the following commands from the archive root:

```
python3 -B direct/code/verify.py
python3 -B -O direct/code/verify.py
python3 -B direct/code/verify.py --rebuild
```

The optional rebuild uses a temporary directory and compares nineteen generated JSON objects byte for byte. It requires no network or repository checkout. All arithmetic checkers use only the Python standard library. The enclosing `README.md` gives the exact native and compact commands and the report's LaTeX compilation command. Source licensing and AI assistance notices are retained in each component.

The result is best described as a concrete improvement in several explicit constructions. It is not established to be the globally simplest Diophantine equation for consistency. In particular, fewer variables alone would favor some different higher-degree constructions, while smaller coefficients favor the direct quartic. The archive keeps those comparisons separate rather than counting one measure as all of them.

10 Further research questions

1. **A systematic common-remainder compiler.** Which digit-based Diophantine constructions have canonical remainders uniformly much smaller than their moduli? Theorem 2.2 applies to any such family, but proving the margin before decoding is the main semantic obligation. The divisors need only share an explicitly certified common multiple.
2. **A better common multiple.** The choice $M = q^x$ is convenient because it already exists. A different grouping or smaller common multiple could reduce support or the number of ratio coordinates. Any saving must include the variables needed to certify the new multiple and its exact quotients.
3. **The three initial sign guards.** The remaining arithmetic bootstrap still uses the initial remainder, slack, and output-slack signs. Replacing them requires a new noncircular proof of $B > 0$ and $q \geq 2$. The common-remainder lemma alone does not justify deleting these guards.

4. **Degree and support after coordinate changes.** The current monic eliminations optimize variable count subject to a sparse substitution cap. Alternative integral coordinates could lower the 5,340-term support or coefficient height while preserving 252 variables. Such improvements should be measured on collected polynomials and accompanied by inverse maps.

5. **Native label choices and carry proofs.** Label names are semantically arbitrary, but their binary values affect selector coefficients and carry bounds. A finite optimization over valid color bijections is potentially useful. It must preserve complete boundary and occupancy semantics, and must prove all downstream resets, including any newly created clause carry. Below radix47, the current uniform clause-predecessor bound $7 \cdot 2^{44} - 2^{10}$ itself exceeds the next radix; a smaller global coefficient sum alone does not resolve that local obstruction.

6. **Formalizing the new local theorem.** Lemma 2.1, the common norm theorem, and Lemma 4.1 are small, independent formalization targets. Connecting their verified source transformation to the complete literal polynomial would provide a focused proof-assistant contribution without claiming to rebuild the entire inherited compiler at once.

References

[1] V. Reshetnikov, *ProveIt*, source snapshot `55d1f72aedebf892e446c90d35fa1ae37689b0f2`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/55d1f72aedebf892e446c90d35fa1ae37689b0f2/Computability/DiophantineConsistency.

[2] ProveIt incoming research package, *Cofinal Anchors and Height One, Round 17*, 25 October 2026, particularly `direct/STRICT_BUDGET_PROOF.md`, `direct/ORIENTATION_FIRST_PROOF.md`, and the `strict_dimension_centered` literal equation. The exact source archive is `docs/incoming/ConZFC_Cofinal_Anchors_Height_One_Round17_2026-10-25.zip` at [1]. Relevant proof texts and exact input objects are retained in this release.

[3] ProveIt incoming research package, *Certificate Reductions*, 25 October 2026, `compact/PROOF.md` and its two fixed 78/630 and 79/468 circuits. Source archive: `docs/incoming/ConZFC_Certificate_Reductions_2026-10-25.zip` at [1].

[4] ProveIt, *Seven shared marker weights and a conditional radix49 adapter*, `research/compact66-shared-marker-radix49/README.md` at [1]. The architecture and its ordinary proof precede the finite certificate completion in this report.

[5] ProveIt, *A radix47 alias of the isolated shared-marker clause test*, `research/compact66-shared-marker-radix47-clause-alias/README.md` at [1]. This is a scoped obstruction to a former local argument, not an impossibility theorem for radix47 encodings.

[6] Y. Matiyasevich, *Hilbert's Tenth Problem*, MIT Press, 1993; author's online preface and commentary: https://logic.pdmi.ras.ru/~yumat/H10Pbook/preface.htm.

[7] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* **47** (1982), 549–571. https://doi.org/10.2307/2273588.

[8] J. Bayer, M. David, M. Hassler, Y. Matiyasevich, and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, 2025. https://arxiv.org/abs/2506.20909. Related checked development: https://isa-afp.org/entries/Diophantine_Universal_Pairs.html.

A The complete direct polynomial

All $z_0, \ldots, z_{251}$ are unrestricted integers. The following is the complete list of fixed residuals. No abbreviation in this appendix is an additional quantified variable. The requested single equation is

$$\boxed{\sum_{j=0}^{181} Q_j(\mathbf{z})^2 = 0.}$$

The authoritative variable order, residual coefficients, and full collected expansion are also supplied in machine-readable form. The appendix is generated directly from that final JSON, and its metadata and coefficient expansion are independently checked.

$$\begin{aligned}
Q_0 &= -z_3^2 + z_4 \\
Q_1 &= z_0 - z_0 z_2 - z_2 + z_9 \\
Q_2 &= -1 + z_1 - z_2 z_7 - z_3 \\
Q_3 &= -1 + z_1 - z_8 z_9 \\
Q_4 &= 1 + z_4 - z_5 z_{20} + z_{18} \\
Q_5 &= -1 + 2z_{10} - z_{10}^2 - 2z_{10}z_{11} + 2z_{11} - z_{11}^2 + z_{21} \\
Q_6 &= -5 - 30z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 24z_{10}z_{21} - 40z_{11} \\
&\quad - 64z_{11}^2 - 32z_{11}z_{21} - 10z_{21} - 4z_{21}^2 + z_{22} \\
Q_7 &= -2 + 2z_{10} - z_{10}^2 - 4z_{10}z_{22} + 4z_{22} - 4z_{22}^2 + z_{23} \\
Q_8 &= 1 + 6z_{10} - 36z_{10}^2 - 192z_{10}z_{22} - 24z_{10}z_{23} + 16z_{22} \\
&\quad - 256z_{22}^2 - 64z_{22}z_{23} + 2z_{23} - 4z_{23}^2 + z_{24} \\
Q_9 &= -2 - z_{11}^2 - 2z_{11}z_{12} - z_{12}^2 + z_{25} \\
Q_{10} &= -9 + 7z_{10} - z_{10}^2 - 4z_{10}z_{11} - 4z_{10}z_{25} + 12z_{11} \\
&\quad - 4z_{11}^2 - 8z_{11}z_{25} + 12z_{25} - 4z_{25}^2 + z_{26} \\
Q_{11} &= -1 + 2z_{10} - z_{10}^2 - 2z_{10}z_{12} + 2z_{12} - z_{12}^2 + z_{27} \\
Q_{12} &= -6z_{10} - 144z_{10}^2 - 192z_{10}z_{11} - 192z_{10}z_{12} \\
&\quad - 48z_{10}z_{21} - 48z_{10}z_{27} - 8z_{11} - 64z_{11}^2 - 128z_{11}z_{12} \\
&\quad - 32z_{11}z_{21} - 32z_{11}z_{27} - 64z_{12}^2 - 32z_{12}z_{21} \\
&\quad - 32z_{12}z_{27} - 2z_{21} - 4z_{21}^2 - 8z_{21}z_{27} - 4z_{27}^2 + z_{28} \\
Q_{13} &= -4z_{26}^2 - 8z_{26}z_{28} - 4z_{28}^2 + z_{29} \\
Q_{14} &= -6 + 16z_{12} - 16z_{12}^2 - 16z_{12}z_{25} - 16z_{12}z_{27} + 8z_{25} \\
&\quad - 4z_{25}^2 - 8z_{25}z_{27} + 8z_{27} - 4z_{27}^2 + z_{30}
\end{aligned}$$

$$
\begin{aligned}
Q_{15} = {} & 1 + 6z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 48z_{10}z_{12} \\
& - 24z_{10}z_{21} - 48z_{10}z_{27} - 24z_{10}z_{30} + 8z_{11} - 64z_{11}^2 \\
& - 64z_{11}z_{12} - 32z_{11}z_{21} - 64z_{11}z_{27} - 32z_{11}z_{30} \\
& + 8z_{12} - 16z_{12}^2 - 16z_{12}z_{21} - 32z_{12}z_{27} - 16z_{12}z_{30} \\
& + 2z_{21} - 4z_{21}^2 - 16z_{21}z_{27} - 8z_{21}z_{30} + 8z_{27} \\
& - 16z_{27}^2 - 16z_{27}z_{30} + 4z_{30} - 4z_{30}^2 + z_{31}
\end{aligned}
$$

$$
\begin{aligned}
Q_{16} = {} & -10 - 5z_{10} - z_{10}^2 - 12z_{10}z_{11} - 16z_{10}z_{12} \\
& - 4z_{10}z_{25} - 36z_{11} - 36z_{11}^2 - 96z_{11}z_{12} - 24z_{11}z_{25} \\
& - 48z_{12} - 64z_{12}^2 - 32z_{12}z_{25} - 12z_{25} - 4z_{25}^2 + z_{32}
\end{aligned}
$$

$$
\begin{aligned}
Q_{17} = {} & -2 - 24z_{10} - 36z_{10}^2 - 12z_{10}z_{11} - 96z_{10}z_{12} \\
& - 24z_{10}z_{27} - 3z_{11} - z_{11}^2 - 16z_{11}z_{12} - 4z_{11}z_{27} \\
& - 32z_{12} - 64z_{12}^2 - 32z_{12}z_{27} - 8z_{27} - 4z_{27}^2 + z_{33}
\end{aligned}
$$

$$
Q_{18} = 2 - 4z_{32}^2 - 8z_{32}z_{33} - 4z_{33}^2 + z_{34}
$$

$$
\begin{aligned}
Q_{19} = {} & -12 + 16z_{10} - 4z_{10}^2 - 8z_{10}z_{12} - 8z_{10}z_{21} \\
& - 8z_{10}z_{25} + 16z_{12} - 4z_{12}^2 - 8z_{12}z_{21} - 8z_{12}z_{25} \\
& + 16z_{21} - 4z_{21}^2 - 8z_{21}z_{25} + 16z_{25} - 4z_{25}^2 + z_{35}
\end{aligned}
$$

$$
\begin{aligned}
Q_{20} = {} & -3 - 24z_{10} - 64z_{10}^2 - 160z_{10}z_{12} - 64z_{10}z_{25} \\
& - 32z_{10}z_{27} - 32z_{10}z_{35} - 26z_{12} - 100z_{12}^2 \\
& - 80z_{12}z_{25} - 40z_{12}z_{27} - 40z_{12}z_{35} - 8z_{25} - 16z_{25}^2 \\
& - 16z_{25}z_{27} - 16z_{25}z_{35} - 6z_{27} - 4z_{27}^2 - 8z_{27}z_{35} \\
& - 4z_{35} - 4z_{35}^2 + z_{36}
\end{aligned}
$$

$$
\begin{aligned}
Q_{21} = {} & -1 - 18z_{10} - 36z_{10}^2 - 96z_{10}z_{11} - 24z_{10}z_{21} - 24z_{11} \\
& - 64z_{11}^2 - 32z_{11}z_{21} - 6z_{21} - 4z_{21}^2 + z_{37}
\end{aligned}
$$

$$
Q_{22} = -2 + 2z_{10} - z_{10}^2 - 4z_{10}z_{37} + 4z_{37} - 4z_{37}^2 + z_{38}
$$

$$
\begin{aligned}
Q_{23} = {} & 1 + 6z_{10} - 36z_{10}^2 - 192z_{10}z_{37} - 24z_{10}z_{38} + 16z_{37} \\
& - 256z_{37}^2 - 64z_{37}z_{38} + 2z_{38} - 4z_{38}^2 + z_{39}
\end{aligned}
$$

$$
Q_{24} = -z_{11}^2 - 4z_{11}z_{39} - 4z_{39}^2 + z_{40}
$$

$$
\begin{aligned}
Q_{25} = {} & -6 - 30z_{10} - 64z_{10}^2 - 128z_{10}z_{11} - 64z_{10}z_{21} - 32z_{11} \\
& - 64z_{11}^2 - 64z_{11}z_{21} - 14z_{21} - 16z_{21}^2 + z_{41}
\end{aligned}
$$

$$
\begin{aligned}
Q_{26} = {} & -4 + 8z_{11} - 64z_{11}^2 - 96z_{11}z_{12} - 32z_{11}z_{25} \\
& - 32z_{11}z_{41} + 6z_{12} - 36z_{12}^2 - 24z_{12}z_{25} - 24z_{12}z_{41} \\
& + 2z_{25} - 4z_{25}^2 - 8z_{25}z_{41} + 4z_{41} - 4z_{41}^2 + z_{42}
\end{aligned}
$$

$$
\begin{aligned}
Q_{27} = {} & 8z_{10} - 64z_{10}^2 - 96z_{10}z_{12} - 32z_{10}z_{27} - 32z_{10}z_{42} \\
& + 6z_{12} - 36z_{12}^2 - 24z_{12}z_{27} - 24z_{12}z_{42} + 2z_{27} \\
& - 4z_{27}^2 - 8z_{27}z_{42} + 4z_{42} - 4z_{42}^2 + z_{43}
\end{aligned}
$$

$$
Q_{28} = -z_{12}^2 - 2z_{12}z_{13} - z_{13}^2 + z_{44}
$$

$$
\begin{aligned}
Q_{29} &= 2z_{10} - z_{10}^2 - 4z_{10}z_{12} - 4z_{10}z_{44} + 4z_{12} - 4z_{12}^2 \\
&\quad - 8z_{12}z_{44} + 4z_{44} - 4z_{44}^2 + z_{45}
\end{aligned}
$$

$$
\begin{aligned}
Q_{30} &= -35 + 24z_{10} - 4z_{10}^2 - 8z_{10}z_{11} - 8z_{10}z_{25} \\
&\quad - 8z_{10}z_{27} + 24z_{11} - 4z_{11}^2 - 8z_{11}z_{25} - 8z_{11}z_{27} \\
&\quad + 24z_{25} - 4z_{25}^2 - 8z_{25}z_{27} + 24z_{27} - 4z_{27}^2 + z_{46}
\end{aligned}
$$

$$
\begin{aligned}
Q_{31} &= -16z_{10}^2 - 32z_{10}z_{12} - 32z_{10}z_{27} - 32z_{10}z_{44} \\
&\quad - 16z_{10}z_{45} - 16z_{10}z_{46} - 4z_{12} - 16z_{12}^2 - 32z_{12}z_{27} \\
&\quad - 32z_{12}z_{44} - 16z_{12}z_{45} - 16z_{12}z_{46} - 16z_{27}^2 \\
&\quad - 32z_{27}z_{44} - 16z_{27}z_{45} - 16z_{27}z_{46} - 4z_{44} - 16z_{44}^2 \\
&\quad - 16z_{44}z_{45} - 16z_{44}z_{46} - 2z_{45} - 4z_{45}^2 - 8z_{45}z_{46} \\
&\quad - 4z_{46}^2 + z_{47}
\end{aligned}
$$

$$
\begin{aligned}
Q_{32} &= z_{12} - 25z_{12}^2 - 40z_{12}z_{27} - 20z_{12}z_{30} - 16z_{27}^2 \\
&\quad - 16z_{27}z_{30} - 4z_{30}^2 + z_{48}
\end{aligned}
$$

$$
Q_{33} = -1 + 8z_{11} - 4z_{11}^2 - 8z_{11}z_{25} + 8z_{25} - 4z_{25}^2 + z_{49}
$$

$$
\begin{aligned}
Q_{34} &= -1 - 16z_{11}^2 - 8z_{11}z_{12} - 32z_{11}z_{25} - 16z_{11}z_{49} \\
&\quad - z_{12}^2 - 8z_{12}z_{25} - 4z_{12}z_{49} - 16z_{25}^2 - 16z_{25}z_{49} \\
&\quad - 4z_{49}^2 + z_{50}
\end{aligned}
$$

$$
\begin{aligned}
Q_{35} &= 1 + z_{11} - z_{11}^2 - 4z_{11}z_{12} - 4z_{11}z_{50} - 4z_{12}^2 \\
&\quad - 8z_{12}z_{50} - 4z_{50}^2 + z_{51}
\end{aligned}
$$

$$
Q_{36} = -1 - 4z_{51} - 4z_{51}^2 + z_{52}
$$

$$
\begin{aligned}
Q_{37} &= -1 - z_{12}^2 - 8z_{12}z_{51} - 4z_{12}z_{52} - 16z_{51}^2 \\
&\quad - 16z_{51}z_{52} - 4z_{52}^2 + z_{53}
\end{aligned}
$$

$$
\begin{aligned}
Q_{38} &= 2 + z_{11} - z_{11}^2 - 4z_{11}z_{12} - 4z_{11}z_{53} - 4z_{12}^2 \\
&\quad - 8z_{12}z_{53} - 4z_{53}^2 + z_{54}
\end{aligned}
$$

$$
Q_{39} = -4z_{48}^2 - 8z_{48}z_{54} - 4z_{54}^2 + z_{55}
$$

$$
\begin{aligned}
Q_{40} &= -9 + 7z_{10} - z_{10}^2 - 8z_{10}z_{48} - 4z_{10}z_{55} + 24z_{48} \\
&\quad - 16z_{48}^2 - 16z_{48}z_{55} + 12z_{55} - 4z_{55}^2 + z_{56}
\end{aligned}
$$

$$
Q_{41} = -1 - 4z_{56} - 4z_{56}^2 + z_{57}
$$

$$
\begin{aligned}
Q_{42} &= -z_{12}^2 - 8z_{12}z_{56} - 4z_{12}z_{57} - 16z_{56}^2 - 16z_{56}z_{57} \\
&\quad - 4z_{57}^2 + z_{58}
\end{aligned}
$$

$$
\begin{aligned}
Q_{43} &= -z_{10} - z_{10}^2 - 4z_{10}z_{12} - 4z_{10}z_{58} - 4z_{12} - 4z_{12}^2 \\
&\quad - 8z_{12}z_{58} - 4z_{58} - 4z_{58}^2 + z_{59}
\end{aligned}
$$

$$
Q_{44} = -10 - 12z_{59} - 4z_{59}^2 + z_{60}
$$

$$
\begin{aligned}
Q_{45} &= 1 + 4z_{11} - 4z_{11}^2 - 8z_{11}z_{25} - 16z_{11}z_{59} - 8z_{11}z_{60} \\
&\quad + 4z_{25} - 4z_{25}^2 - 16z_{25}z_{59} - 8z_{25}z_{60} + 8z_{59} \\
&\quad - 16z_{59}^2 - 16z_{59}z_{60} + 4z_{60} - 4z_{60}^2 + z_{61}
\end{aligned}
$$

$$
\begin{aligned}
Q_{46} &= z_{11} - 25z_{11}^2 - 40z_{11}z_{25} - 20z_{11}z_{61} - 16z_{25}^2 \\
&\quad - 16z_{25}z_{61} - 4z_{61}^2 + z_{62} \\
Q_{47} &= -10 - 12z_{54} - 4z_{54}^2 + z_{63} \\
Q_{48} &= 1 - 16z_{54}^2 - 16z_{54}z_{62} - 16z_{54}z_{63} - 4z_{62}^2 \\
&\quad - 8z_{62}z_{63} - 4z_{63}^2 + z_{64} \\
Q_{49} &= z_{12} - z_{12}^2 - 8z_{12}z_{62} - 4z_{12}z_{64} - 16z_{62}^2 \\
&\quad - 16z_{62}z_{64} - 4z_{64}^2 + z_{65} \\
Q_{50} &= -4z_{10} - 4z_{10}^2 - 4z_{10}z_{12} - 8z_{10}z_{21} - z_{12} - z_{12}^2 \\
&\quad - 4z_{12}z_{21} - 4z_{21} - 4z_{21}^2 + z_{66} \\
Q_{51} &= 1 + 4z_{10} - 4z_{10}^2 - 8z_{10}z_{21} - 8z_{10}z_{66} + 4z_{21} \\
&\quad - 4z_{21}^2 - 8z_{21}z_{66} + 4z_{66} - 4z_{66}^2 + z_{67} \\
Q_{52} &= -2 - 4z_{10} - 16z_{10}^2 - 32z_{10}z_{21} - 16z_{10}z_{65} \\
&\quad - 16z_{10}z_{67} - 4z_{21} - 16z_{21}^2 - 16z_{21}z_{65} - 16z_{21}z_{67} \\
&\quad - 4z_{65}^2 - 8z_{65}z_{67} - 2z_{67} - 4z_{67}^2 + z_{68} \\
Q_{53} &= -2 - 4z_{68} - 4z_{68}^2 + z_{69} \\
Q_{54} &= -z_3z_{15} + z_{70} \\
Q_{55} &= z_{14} - z_{15} - 2z_{24}z_{70} - z_{71} \\
Q_{56} &= z_{14} - z_{15} - 4z_{26}z_{70} - 2z_{29}z_{70} - z_{72} \\
Q_{57} &= z_{14} - z_{15} - 2z_{31}z_{70} - 6z_{70} - z_{73} \\
Q_{58} &= z_{14} - z_{15} - 4z_{32}z_{70} - 2z_{34}z_{70} - z_{74} \\
Q_{59} &= z_{14} - z_{15} - 2z_{36}z_{70} + 6z_{70} - z_{75} \\
Q_{60} &= z_{14} - z_{15} - 4z_{39}z_{70} - 2z_{40}z_{70} - z_{76} \\
Q_{61} &= z_{14} - z_{15} - 2z_{43}z_{70} - z_{77} \\
Q_{62} &= z_{14} - z_{15} - 2z_{47}z_{70} - 4z_{70} - z_{78} \\
Q_{63} &= -z_{10}z_{11} + z_{10}z_{12} + z_{11} + z_{11}z_{12} - z_{12} \\
&\quad - z_{12}^2 + z_{79} \\
Q_{64} &= 1 + 2z_{13} + z_{13}^2 - z_{79}^2 + z_{80} \\
Q_{65} &= -z_{80}^2 + z_{81} \\
Q_{66} &= -4z_{68}z_{70} - 2z_{69}z_{70} + 2z_{70} + z_{82} \\
Q_{67} &= -z_{14}^2 + 2z_{14}z_{15} + 2z_{14}z_{82} - z_{15}^2 - 2z_{15}z_{82} \\
&\quad - z_{82}^2 + z_{83} \\
Q_{68} &= -1 - z_3z_{11} + z_{84} \\
Q_{69} &= -z_{15}z_{84} + z_{86}z_{87} + z_{88} \\
Q_{70} &= -1 + z_3 - z_3z_{10} + z_{85} \\
Q_{71} &= -1 + 2z_3 - 2z_3z_{10} - 2z_3z_{21} + z_{86}
\end{aligned}
$$

$$
\begin{aligned}
Q_{72} &= -z_{14}z_{85} + z_{87} \\
Q_{73} &= -1 - 6z_3z_{10} - 8z_3z_{11} - 2z_3z_{21} + z_{89} \\
Q_{74} &= z_{14}z_{84} - z_{15}z_{89} + z_{90} \\
Q_{75} &= -z_{14}z_{71} + z_{15}z_{71} + z_{91} \\
Q_{76} &= -z_{72}z_{91} + z_{92} \\
Q_{77} &= -z_{73}z_{92} + z_{93} \\
Q_{78} &= -z_{74}z_{93} + z_{94} \\
Q_{79} &= -z_{75}z_{94} + z_{95} \\
Q_{80} &= -z_{76}z_{95} + z_{96} \\
Q_{81} &= -z_{77}z_{96} + z_{97} \\
Q_{82} &= -z_{78}z_{97} + z_{98} \\
Q_{83} &= -z_{88}z_{98} + z_{99} \\
Q_{84} &= -z_{90}z_{99} + z_{100} \\
Q_{85} &= -z_{100}^2 + z_{101} \\
Q_{86} &= -z_2 - z_2z_{18} + z_{18} + z_{102} \\
Q_{87} &= -z_{16}z_{102} + z_{103} \\
Q_{88} &= -2z_1^2 + 2z_1z_{14} + 2z_1z_{15} + 2z_1z_{103} + 2z_1z_{104} \\
&\quad - z_{14}^2 - 2z_{14}z_{103} - z_{15}^2 - 2z_{15}z_{104} + z_{19}z_{106} \\
&\quad - z_{81}z_{101} - z_{83}z_{101} - z_{103}^2 - z_{104}^2 \\
Q_{89} &= -z_2z_{17} + z_{17} - z_{17}z_{102} + z_{104} \\
Q_{90} &= -9z_{10} - 9z_{11} - 4z_{21} - 2z_{24} - 4z_{26} - 2z_{29} - 2z_{31} \\
&\quad - 4z_{32} - 2z_{34} - 2z_{36} - 4z_{39} - 2z_{40} - 2z_{43} - 2z_{47} \\
&\quad - 4z_{68} - 2z_{69} + z_{105} \\
Q_{91} &= -z_{107}z_{110} + z_{110} + z_{112} \\
Q_{92} &= -z_5z_{134} + z_{108}z_{234} - z_{111} - z_{112}z_{134} \\
Q_{93} &= z_{113}z_{235} - z_{115}z_{137} - z_{116} \\
Q_{94} &= z_{117}z_{235} - z_{118}z_{137} - z_{119} \\
Q_{95} &= z_{120}z_{235} - z_{121}z_{137} - z_{122} \\
Q_{96} &= z_{123}z_{235} - z_{124}z_{137} - z_{125} \\
Q_{97} &= z_{126}z_{236} - z_{128}z_{137} - z_{129} \\
Q_{98} &= z_{130}z_{236} - z_{131}z_{137} - z_{132} \\
Q_{99} &= -z_{19}z_{134} + z_{137} \\
Q_{100} &= z_{133} - z_{135}z_{137} - z_{136} \\
Q_{101} &= -z_{137}z_{139} + z_{138} - z_{140} \\
Q_{102} &= -z_{137}z_{142} + z_{141}z_{234} - z_{143}
\end{aligned}
$$

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\begin{aligned}
Q_{103} &= -z_{137}z_{145} + z_{144}z_{235} - z_{146} \\
Q_{104} &= z_{147} - 2z_{147}z_{149} + z_{150} \\
Q_{105} &= -z_{147}z_{150} - z_{150} + z_{151} \\
Q_{106} &= 2z_{148} + z_{148}^2 + z_{150}^2 - z_{151}^2 \\
Q_{107} &= -z_{148}z_{153} - z_{153} + z_{154} \\
Q_{108} &= 2z_{152} + z_{152}^2 + z_{153}^2 - z_{154}^2 \\
Q_{109} &= z_4^2 - 2z_4z_{148} - 2z_{148} + z_{155} \\
Q_{110} &= -1 - z_4z_{153} + z_6 - z_{152} - z_{153} + z_{154} \\
&\quad + z_{155}z_{156} \\
Q_{111} &= 1 - 2z_5 + z_5^2 - 2z_5z_{148} + z_{157} \\
Q_{112} &= -z_5z_{203} + z_{19} + z_{157}z_{158} - z_{202} + z_{204} \\
Q_{113} &= -1 + 2z_{148}z_{218} - z_{159} + 2z_{218} - z_{218}^2 \\
Q_{114} &= -1 + z_{107} + z_{159}z_{160} - z_{202} - z_{203}z_{218} + z_{204} \\
Q_{115} &= -z_{148}z_{162} - z_{162} + z_{163} \\
Q_{116} &= 2z_{161} + z_{161}^2 + z_{162}^2 - z_{163}^2 \\
Q_{117} &= 1 - 2z_{107} + z_{107}^2 - 2z_{107}z_{148} + z_{164} \\
Q_{118} &= -1 - z_{107}z_{162} + z_{108} - z_{161} + z_{163} + z_{164}z_{165} \\
Q_{119} &= 4 - 4z_{107} + z_{107}^2 - 2z_{107}z_{148} + 2z_{148} + z_{166} \\
Q_{120} &= -1 - z_{107}z_{153} + z_{109} - z_{152} + z_{153} + z_{154} \\
&\quad + z_{166}z_{167} \\
Q_{121} &= 1 + z_{148} - z_{148}z_{169} - z_{169} + z_{170} \\
Q_{122} &= 1 + 2z_{168} + z_{168}^2 - 2z_{169} + z_{169}^2 - z_{170}^2 \\
Q_{123} &= -1 + z_{107} - z_{107}z_{169} + z_{113} + z_{164}z_{171} - z_{168} \\
&\quad + z_{170} \\
Q_{124} &= -2z_{148} - z_{148}^2 + z_{172} \\
Q_{125} &= 1 - z_{107} - z_{107}z_{152} - z_{107}z_{154} + z_{114} + z_{148}z_{154} \\
&\quad + z_{152} - z_{153}z_{172} + 2z_{154} + z_{166}z_{173} \\
Q_{126} &= -z_{148}z_{175} - z_{175} + z_{176} \\
Q_{127} &= 2z_{174} + z_{174}^2 + z_{175}^2 - z_{176}^2 \\
Q_{128} &= -1 - z_{107}z_{175} + z_{117} + z_{164}z_{177} - z_{174} + z_{176} \\
Q_{129} &= -z_{148}z_{179} - z_{179} + z_{180} \\
Q_{130} &= 2z_{178} + z_{178}^2 + z_{179}^2 - z_{180}^2 \\
Q_{131} &= -1 - z_{107}z_{179} + z_{120} + z_{164}z_{181} - z_{178} + z_{180} \\
Q_{132} &= -z_{148}z_{183} - z_{183} + z_{184} \\
Q_{133} &= 2z_{182} + z_{182}^2 + z_{183}^2 - z_{184}^2
\end{aligned}
$$

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\begin{aligned}
Q_{134} = & -1 - z_{107}z_{183} + z_{123} + z_{164}z_{185} - z_{182} + z_{184} \\
Q_{135} = & -z_{148}z_{187} - z_{187} + z_{188} \\
Q_{136} = & \, 2z_{186} + z_{186}^2 + z_{187}^2 - z_{188}^2 \\
Q_{137} = & -1 - z_{107}z_{187} + z_{126} + z_{164}z_{189} - z_{186} + z_{188} \\
Q_{138} = & -z_{148}z_{191} - z_{191} + z_{192} \\
Q_{139} = & \, 2z_{190} + z_{190}^2 + z_{191}^2 - z_{192}^2 \\
Q_{140} = & -1 - z_{107}z_{191} + z_{127} + z_{166}z_{193} - z_{190} + z_{191} \\
& + z_{192} \\
Q_{141} = & -z_{148}z_{195} - z_{195} + z_{196} \\
Q_{142} = & \, 2z_{194} + z_{194}^2 + z_{195}^2 - z_{196}^2 \\
Q_{143} = & -1 - z_{107}z_{195} + z_{130} + z_{164}z_{197} - z_{194} + z_{196} \\
Q_{144} = & -z_{148}z_{199} - z_{199} + z_{200} \\
Q_{145} = & \, 2z_{198} + z_{198}^2 + z_{199}^2 - z_{200}^2 \\
Q_{146} = & -1 - z_{107}z_{199} + z_{133} + z_{164}z_{201} - z_{198} + z_{200} \\
Q_{147} = & \, 1 - z_{148}z_{203} - z_{203} + z_{204} \\
Q_{148} = & -1 + 2z_{202} + z_{202}^2 + z_{203}^2 - 2z_{204} - z_{204}^2 \\
Q_{149} = & -z_{107}z_{203} + z_{134} + z_{166}z_{205} - z_{202} + z_{203} + z_{204} \\
Q_{150} = & -z_{148}z_{207} - z_{207} + z_{208} \\
Q_{151} = & \, 2z_{206} + z_{206}^2 + z_{207}^2 - z_{208}^2 \\
Q_{152} = & -1 - z_{107}z_{207} + z_{138} + z_{164}z_{209} - z_{206} + z_{208} \\
Q_{153} = & -z_{148}z_{211} - z_{211} + z_{212} \\
Q_{154} = & \, 2z_{210} + z_{210}^2 + z_{211}^2 - z_{212}^2 \\
Q_{155} = & -1 - z_{107}z_{211} + z_{141} + z_{164}z_{213} - z_{210} + z_{212} \\
Q_{156} = & -z_{148}z_{215} - z_{215} + z_{216} \\
Q_{157} = & \, 2z_{214} + z_{214}^2 + z_{215}^2 - z_{216}^2 \\
Q_{158} = & -1 - z_{107}z_{215} + z_{144} + z_{164}z_{217} - z_{214} + z_{216} \\
Q_{159} = & -6 - 2z_0 - 3z_0^2 - 2z_0z_4 - z_1^2 - z_2^2 - 2z_4 \\
& - 2z_4^2 - z_5^2 - z_6^2 + 2z_{10} - z_{10}^2 - z_{11}^2 - z_{12}^2 \\
& - z_{13}^2 - z_{14}^2 - z_{15}^2 - z_{16}^2 - z_{17}^2 - z_{18}^2 - z_{19}^2 \\
& - z_{105}^2 + 2z_{107} - 2z_{107}^2 - z_{108}^2 - z_{109}^2 - z_{113}^2 - z_{114}^2 \\
& - z_{117}^2 - z_{120}^2 - z_{123}^2 - z_{126}^2 - z_{127}^2 - z_{130}^2 - z_{133}^2 \\
& - z_{134}^2 - z_{138}^2 - z_{141}^2 - z_{144}^2 + z_{147} - z_{218}^2 \\
Q_{160} = & -z_5z_{10} - z_5z_{11} - z_5z_{12} - z_5z_{13} \\
& - z_5z_{14} - z_5z_{15} - z_5z_{16} - z_5z_{17} \\
& - z_5z_{18} + z_{218}
\end{aligned}
$$

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\begin{aligned}
Q_{161} =& -16 + z_0 - z_{219}^2 \\
Q_{162} =& -1 + z_2 + z_6 - z_6^2 \\
Q_{163} =& -z_{148}z_{221} - z_{221} + z_{230} \\
Q_{164} =& -1 + z_{220}^2 + z_{221}^2 - z_{230}^2 \\
Q_{165} =& -z_{223} - z_{223}z_{224} + z_{231} \\
Q_{166} =& -1 + z_{222}^2 + z_{223}^2 - z_{231}^2 \\
Q_{167} =& -z_{215}^2 + z_{232} \\
Q_{168} =& z_{221} - 2z_{226}z_{232} - z_{232} \\
Q_{169} =& -4z_{221} - 8z_{221}z_{225} + z_{224} \\
Q_{170} =& -z_{148} - z_{220}z_{227} + z_{224} \\
Q_{171} =& -1 - z_{214} - z_{220}z_{228} + z_{222} \\
Q_{172} =& -z_{105} - 4z_{215}z_{229} + z_{223} \\
Q_{173} =& 2z_0^2 + 2z_0z_4 - 2z_0z_{153} - 2z_0z_{162} + z_1^2 \\
& - 2z_1z_{203} + z_2^2 - 2z_2z_{191} + z_4^2 - 2z_4z_{162} \\
& + z_{10}^2 - 2z_{10}z_{169} + z_{11}^2 - 2z_{11}z_{175} + z_{12}^2 \\
& - 2z_{12}z_{179} + z_{13}^2 - 2z_{13}z_{183} + z_{14}^2 - 2z_{14}z_{187} \\
& + z_{15}^2 - 2z_{15}z_{195} + z_{16}^2 - 2z_{16}z_{199} + z_{17}^2 \\
& - 2z_{17}z_{207} + z_{18}^2 - 2z_{18}z_{211} + z_{105}^2 - 2z_{105}z_{215} \\
& - z_{148}z_{233} + z_{152}^2 + z_{153}^2 + z_{161}^2 + z_{162}^2 + z_{168}^2 \\
& + z_{169}^2 + z_{174}^2 + z_{175}^2 + z_{178}^2 + z_{179}^2 + z_{182}^2 + z_{183}^2 \\
& + z_{186}^2 + z_{187}^2 + z_{190}^2 + z_{191}^2 + z_{194}^2 + z_{195}^2 + z_{198}^2 \\
& + z_{199}^2 + z_{202}^2 + z_{203}^2 + z_{206}^2 + z_{207}^2 + z_{210}^2 + z_{211}^2 \\
& + z_{214}^2 + z_{215}^2 \\
Q_{174} =& -z_{109}z_{234} + z_{134} \\
Q_{175} =& -z_{114}z_{235} + z_{134} \\
Q_{176} =& -z_{127}z_{236} + z_{134} \\
Q_{177} =& -1 - z_3z_5 + z_5 + z_6 - z_{237}^2 - z_{238}^2 - z_{239} \\
& - z_{239}^2 \\
Q_{178} =& z_3z_5 - z_6 - z_{240}^2 - z_{241}^2 - z_{242} - z_{242}^2 \\
Q_{179} =& -2 - z_5 + z_{107} - z_{243}^2 - z_{244}^2 - z_{245} - z_{245}^2 \\
Q_{180} =& -2 - z_{105}^2 - z_{153}^2 - z_{162}^2 + 2z_{169} - z_{169}^2 - z_{175}^2 \\
& - z_{179}^2 - z_{183}^2 - z_{187}^2 - z_{191}^2 - z_{195}^2 - z_{199}^2 - z_{203}^2 \\
& - z_{207}^2 - z_{211}^2 + z_{215}^2 - z_{246}^2 - z_{247}^2 - z_{248} - z_{248}^2 \\
Q_{181} =& -1 - 4z_{111}^2 - 4z_{116}^2 - 4z_{119}^2 - 4z_{122}^2 - 4z_{125}^2 - 4z_{129}^2 \\
& - 4z_{132}^2 + z_{134}^2 - 4z_{136}^2 - 4z_{140}^2 - 4z_{143}^2 - 4z_{146}^2 - z_{249}^2 \\
& - z_{250}^2 - z_{251} - z_{251}^2
\end{aligned}
$$